CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Advanced Subsidiary Level and GCE Advanced Level

MARK SCHEME for the October/November 2012 series

9702 PHYSICS

9702/23 Paper 2 (AS Structured Questions), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge will not enter into discussions about these mark schemes.

Cambridge is publishing the mark schemes for the October/November 2012 series for most IGCSE, GCE Advanced Level and Advanced Subsidiary Level components and some Ordinary Level components.

CAMBRIDGE
International Examinations

1	**(a)** spacing = 380 or 3.8×10^2 pm	B1	[1]
	(b) time = 24 × 3600 time = 0.086 (0.0864) Ms	B1	[1]
	(c) time = distance / speed = $\dfrac{1.5 \times 10^{11}}{3 \times 10^8}$	C1	
	= 500 (s) = 8.3 min	A1	[2]
	(d) momentum and weight	B1	[1]
	(e) **(i)** arrow to the right of plane direction (about 4° to 24°)	B1	[1]
	(ii) scale diagram drawn or use of cosine formula $v^2 = 250^2 + 36^2 - 2 \times 250 \times 36 \times \cos 45°$ or resolving $v = [(36 \cos 45°)^2 + (250 - 36 \sin 45°)^2]^{1/2}$	C1	
	resultant velocity = 226 (220 – 240 for scale diagram) m s^{-1} allow one mark for values 210 to 219 or 241 to 250 m s^{-1} or use of formula (v^2 = 51068) v = 230 (226) m s^{-1}	A1	[2]
2	**(a)** **(i)** accelerations (A to B and B to C) are same magnitude	B1	
	accelerations (A to B and B to C) are opposite directions or both accelerations are toward B	B1	
	(A to B and B to C) the component of the weight down the slope provides the acceleration	B1	[3]
	(ii) acceleration = $g \sin 15°$	C1	
	$s = 0 + ½ at^2$ $\quad s = 0.26 / \sin 15° = 1.0$	C1	
	$t^2 = \dfrac{1.0 \times 2}{9.8 \times \sin 15°}$ $\quad t = 0.89$ s	A1	[3]
	(iii) $v = 0 + g \sin 15 t$ or $v^2 = 0 + 2g \sin 15 \times 1.0$	C1	
	$v = 2.26 \text{ m s}^{-1}$ (using loss of GPE = gain KE can score full marks)	A1	[2]
	(b) loss of GPE at A = gain in GPE at C or loss of KE at B = gain in GPE at C	B1	
	$h_1 = h_2 = 0.26$ m or $½ mv^2 = mgh$ $\quad h_2 = 0.5 \times (2.26)^2 / 9.81 = 0.26$ m $x = 0.26 / \sin 30° = 0.52$ m	A1	[2]
3	**(a)** power is the rate of doing work or power = work done / time (taken) or power = energy transferred / time (taken)	B1	[1]
	(b) **(i)** as the speed increases drag / air resistance increases	B1	
	resultant force reduces hence acceleration is less	B1	
	constant speed when resultant force is zero (allow one mark for speed increases and acceleration decreases)	B1	[3]

(ii)	force from cyclist = drag force / resistive force	B1	
	$P = 12 \times 48$	M1	
	$P = 576\,\text{W}$	A0	[2]
(iii)	tangent drawn at speed = $8.0\,\text{m s}^{-1}$	M1	
	gradient values that show acceleration between 0.44 to $0.48\,\text{m s}^{-2}$	A1	[2]
(iv)	$F - R = ma$	C1	
	$600 / 8 - R = 80 \times 0.5$ [using $P = 576$] $576 / 8 - R = 80 \times 0.5$	C1	
	$R = 75 - 40 = 35\,\text{N}$ $R = 72 - 40 = 32\,\text{N}$	A1	[3]
(v)	at $12\,\text{m s}^{-1}$ drag is 48 N, at $8\,\text{m s}^{-1}$ drag is 35 or 32 N		
	R / v calculated as 4 and 4 or 4.4		
	and consistent response for whether R is proportional to v or not	B1	[1]

4 (a)	e.m.f. = chemical energy to electrical energy	M1	
	p.d. = electrical energy to thermal energy	M1	
	idea of per unit charge	A1	[3]
(b)	$E = I(R + r)$ or $I = E / (R + r)$ (any subject)	B1	[1]
(c) (i)	$E = 5.8\,\text{V}$	B1	[1]
(ii)	evidence of gradient calculation or calculation with values from graph		
	e.g. $5.8 = 4 + 1.0 \times r$	C1	
	$r = 1.8\,\Omega$	A1	[2]
(d) (i)	$P = VI$	C1	
	$P = 2.9 \times 1.6 = 4.6\ (4.64)\,\text{W}$	A1	[2]
(ii)	power from battery = $1.6 \times 5.8 = 9.28$ or efficiency = VI / EI	C1	
	efficiency = $(4.64 / 9.28) \times 100 = 50\,\%$ or $(2.9 / 5.8) \times 100 = 50\%$	A1	[2]

5 (a)	travel through a vacuum / free space			B1	[1]
(b) (i)	B : name:	**microwaves**	wavelength: 10^{-4} to $10^{-1}\,\text{m}$	B1	
	C : name:	**ultra-violet / UV**	wavelength: 10^{-7} to $10^{-9}\,\text{m}$	B1	
	F : name:	**X –rays**	wavelength: 10^{-9} to $10^{-12}\,\text{m}$	B1	[3]
(ii)	$f = \dfrac{3 \times 10^{8}}{500 \times 10^{-9}}$			C1	
	$f = 6(.0) \times 10^{14}\ \text{Hz}$			A1	[2]

(c)	vibrations are in one direction	M1	
	perpendicular to direction of propagation / energy transfer or good sketch showing this	A1	[2]
6 (a) (i)	electron	B1	[1]
(ii)	any **two:** can be deflected by electric and magnetic fields or negatively charged / absorbed by few (1 – 4) mm of aluminum / 0.5 to 2 m or metres for range in air / speed up to 0.99c / range of speeds / energies	B2	[2]
(iii)	decay occurs and cannot be affected by external / environmental factors or two stated factors such as chemical / pressure / temperature / humidity	B1	[1]
(b)	3 and 0 for superscript numbers	B1	
	2 and –1 for subscript numbers	B1	[2]
(c)	energy = $5.7 \times 10^{3} \times 1.6 \times 10^{-19}$ (= 9.12×10^{-16} J)	C1	
	$v^2 = \dfrac{2 \times 9.12 \times 10^{-16}}{9.11 \times 10^{-31}}$	C1	
	$v = 4.5 \times 10^{7}\ \mathrm{m\,s^{-1}}$	A1	[3]
(d)	both have 1 proton and 1 electron	B1	
	1 neutron in hydrogen-2 and 2 neutrons in hydrogen-3 (special case: for one mark 'same number of protons / atomic number different number of neutrons')	B1	[2]